

# Depot

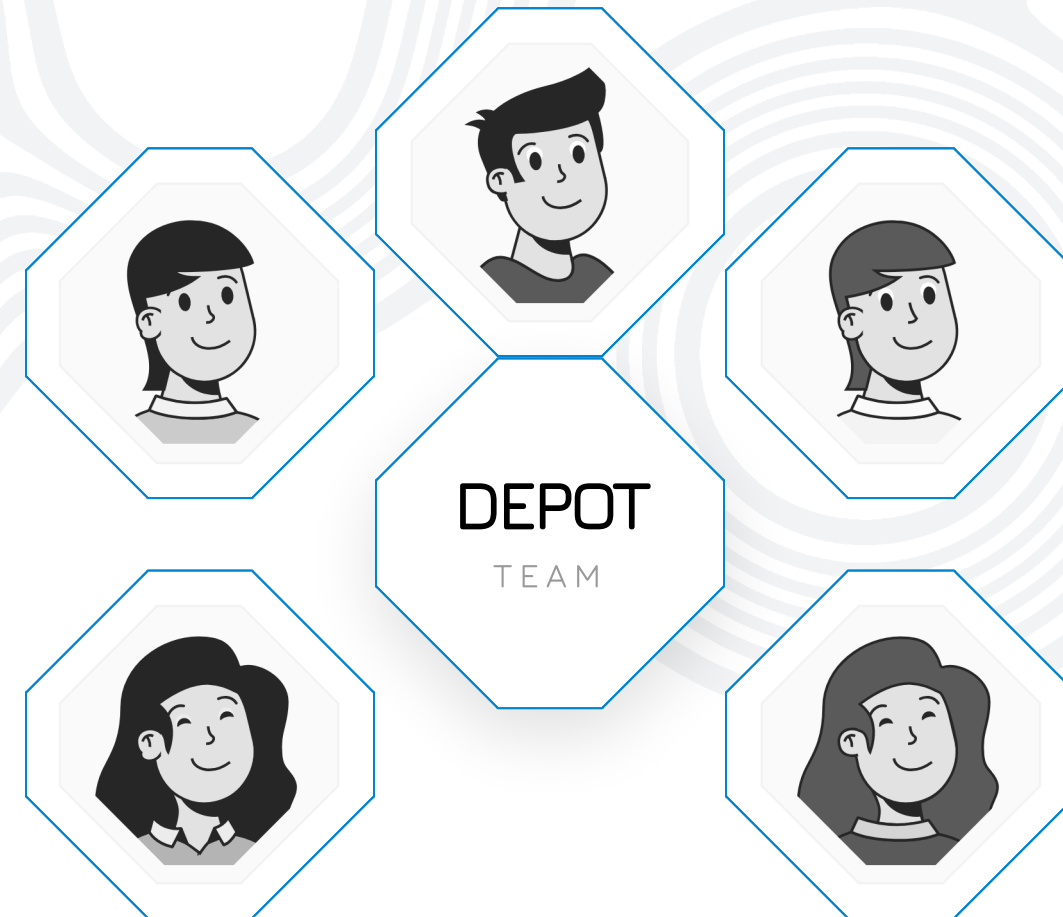
An infrastructure architect powered by AI,  
from code to cloud.

## TEAM

Ahsan Riaz  
Muhammad Waleed  
Muhammad Faizan Anwar

## ADVISORS

Hira Anwar  
Ayesha Hakim



# Building apps is fast now. Shipping them is the bottleneck.

• Building is fast

## ~95%

of the code written by AI, at a quarter of YC's Winter 2025 start-ups

[TechCrunch, 2025](#) [1]

• Deploys break

## ~70%

of AI-written infrastructure failed to deploy on the first try, even from the best of six AI models

[Zhang et al., FSE 2026](#) [2]

• Costs run away

## 29%

of their cloud spend is wasted, say 753 cloud decision-makers

[Flexera State of the Cloud, 2026](#) [3]

• Few can ship it

## 17.8%

of developers work extensively with Terraform, against 43.3% on AWS

[Stack Overflow Developer Survey, 2025](#) [4]

# Working code still leaves four decisions.

Decision 01



What it needs

Decision 02



Where it runs

Decision 03



What it costs

Decision 04



How to change it  
safely

Depot answers all four, **and explains, records and approves every change.**

# Each tool solves part of it.

## Someone still decides where it runs.

|                            | <div>Today</div> <div>Managed platforms</div> <div>Vercel, Railway, Render</div> | <div>Today</div> <div>Bring-your-own-cloud</div> <div>Porter, Qovery, Northflank, Ravion</div> | <div>Today</div> <div>Infrastructure-from-code</div> <div>Encore, Nitric, SST, Defang</div> | <div>Today</div> <div>AI for one provider or codebase</div> <div>Pulumi Neo, StackGen, Gemini Cloud Assist, azd init</div> | <div>Proposed</div> <div>Depot</div> <div>AWS, GCP, Azure, Vercel, Railway; one gate for all</div> |
|----------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Starts from code as it is  | ✓ Yes                                                                            | ⊖ Partly                                                                                       | ✗ No                                                                                        | ⊖ Partly                                                                                                                   | ✓ Yes                                                                                              |
| Runs in the user's account | ✗ No                                                                             | ✓ Yes                                                                                          | ✓ Yes                                                                                       | ⊖ Partly                                                                                                                   | ✓ Yes                                                                                              |
| Chooses cloud or platform  | ✗ No                                                                             | ✗ No                                                                                           | ✗ No                                                                                        | ⊖ Partly                                                                                                                   | ✓ Yes                                                                                              |
| Approval before change     | ⊖ Partly                                                                         | ⊖ Partly                                                                                       | ✗ No                                                                                        | ⊖ Partly                                                                                                                   | ✓ Yes                                                                                              |

Fifteen products, from their own documentation, 30 Sep 2026 [5]. Partly = partly, or only some products; No = no, or not stated. Depot is planned.



# Built for builders without an infrastructure engineer.

• User 01



## AI-assisted builders

A working product, but little experience of running one.

• User 02



## Solo full-stack developers

Outgrowing a managed platform: stay, or move to a cloud?

• User 03



## Small teams and agencies

Two to ten people, with no DevOps staff.

# Three objectives.

## What a user can do with Depot.

01



### Deploy any supported repository

Point Depot at a repo. It reads the app, picks the target, prices it, provisions it.

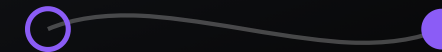
02



### Check health, metrics and cost

History, health, logs, spend and rollback in one place.

03



### Move a live service, not just its code

Cutover planned, priced and rehearsed before the switch.

Research sits inside all three: comparing language models on extraction, and measured guardrails for the agent loop.

# What Depot actually does.

Feature 01



## Application Model

Every fact marked detected, inferred or user-supplied.

Feature 02



## Architect

Picks among reviewed patterns, ranked by cost and effort.

Feature 03



## Safety gate

Validate, saved plan, policy, cost and role check.

Feature 04



## No stored keys

Bounded roles, federation, or one scoped token per target.

Feature 05



## Delivery and operations

Deploy on push, rollback, logs, cost, alerts, savings.

Feature 06



## Agent and incident loop

Changes by conversation; alarms end in a fix a person approves.

# One session, from working code to live.

Step 1



## Connect

GitHub, GitLab or an  
archive

Step 2



## Answer

Up to eight questions

Step 3



## Review

Target, architecture,  
monthly cost

Step 4



## Approve

Signed and role-  
checked

Step 5



## Run

Deploy on push, roll  
back, watch cost

# What Depot covers.

● In scope



GitHub, GitLab or  
an archive



Apps, APIs, static  
sites, workers



AWS, GCP, Azure



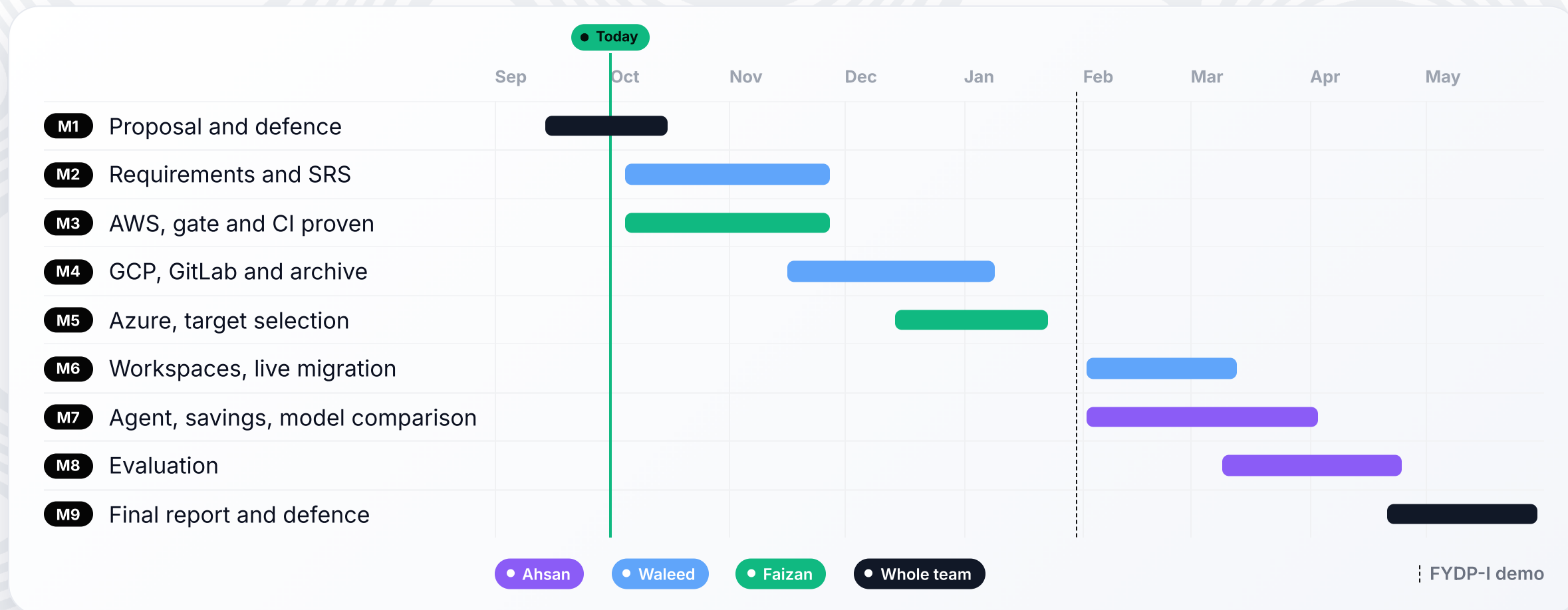
Short-lived  
credentials



Approval before  
every change

Extended scope only if M8 is met early: guided mode, Oracle and Yandex Cloud, voice, preview environments.

# Three clouds first, then everything else.



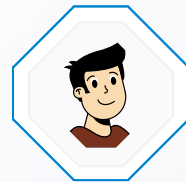
Team planning targets; dates will follow the official defence schedule.



# The Architects

Three owners. No member reviews only their own work.

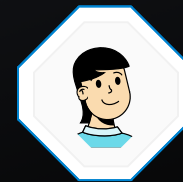
● 01 · 02 · 03



**Ahsan Riaz**

Application Model, agent mode, incident loop, research

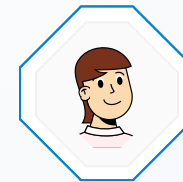
● 01 · 02



**Muhammad Waleed**

User-facing work, team workspaces, managed platforms

● 01 · 03



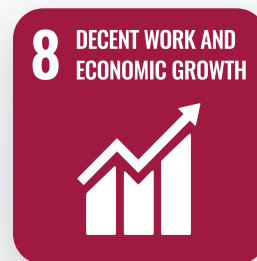
**Muhammad Faizan Anwar**

Architect, cloud adapters, safety gate, savings

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim**

# Two goals we contribute to.

● SDG 8.3



## Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.

● SDG 9.5



## Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

# Sources.

1

[I. Mehta, "A quarter of startups in YC's current cohort have codebases that are almost entirely AI-generated," TechCrunch, Mar. 2025.](#)

2

[T. Zhang, S. Pan, Z. Zhang, Z. Xing and X. Sun, "Deployability-centric infrastructure-as-code generation: Fail, learn, refine, and succeed through LLM-empowered DevOps simulation," ACM FSE, 2026 \(arXiv:2506.05623\).](#)

3

[Flexera, 2026 State of the Cloud Report, 2026.](#)

4

[Stack Overflow, "Technology," 2025 Stack Overflow Developer Survey, 2025.](#)

5

[Vendor documentation of the fifteen products compared, accessed Sep. 30, 2026.](#)

6

[OWASP Gen AI Security Project, "LLM06:2025 Excessive agency," 2025.](#)

7

[United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals, accessed Sep. 30, 2026.](#)

Click a source to open it. Full IEEE entries, with access dates, are in the proposal.

# Thank you. Any questions?

## • TEAM

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